

INTERDISCIPLINARY RESEARCH ARTICLE

Hyper-Ribbon Manifolds of Prediction Error

*Universal Low-Dimensional Structure in Parameterized Models
for 15 Cubic Metallic Elements, with Ecological Fallacy Detection*

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github.com/alexwelcing/lupine | MIT License

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ABSTRACT

Parameterized models in science and engineering are typically evaluated by predictive accuracy on benchmarks. The geometric structure of the errors themselves is less explored. We report a systematic study of 559 independently parameterized models predicting elastic constants (C_{11} , C_{12} , C_{44}) of **15 cubic metallic elements** (8 FCC, 7 BCC). We find that prediction errors universally occupy low-dimensional hyper-ribbon manifolds with effective dimensionality 1.05–1.86 in a 3D observable space — the same geometric signature exhibited by sloppy models in parameter space, here observed for the first time in observable-space error covariance. This compression persists across four decades of model development.

A striking crystal-structure dichotomy governs error correlations: all BCC metals show strong reference–prediction agreement ($r > 0.70$), while FCC metals show near-zero correlation ($r < 0.40$). Random-effects meta-analysis across 15 elements ($N = 1,677$) reveals extreme heterogeneity ($I^2 = 98.6\%$), demonstrating that naive cross-element aggregation commits ecological fallacy. An exploratory analysis of the d -band hypothesis finds it *refuted* on the full sample; the dominant confounder is fitting-sample size rather than electronic structure. A proof-of-concept extension to three foundation MLIPs (MACE-MP-0, CHGNet, Orb-v3) shows error-geometry analysis *can* be applied to neural network potentials, but cross-paradigm universality requires larger samples. We close with a prioritized research roadmap. All data and code are open-source.

Keywords: sloppy models, information geometry, meta-analysis, Simpson's paradox, interatomic potentials, ecological fallacy, machine learning

Data and code: github.com/alexwelcing/lupine (MIT License)

1. Introduction

Every parameterized model produces prediction errors. Whether the model is a neural network, a differential equation, or an analytical potential, the errors carry geometric structure that encodes information about the model, the data, and the underlying physics. Understanding this structure is essential for model improvement, uncertainty quantification, and ensemble design.

We study the geometry of prediction errors in interatomic potentials — mathematical models that describe atomic-scale energy landscapes. Thousands have been developed over forty years, providing an ideal natural experiment. We ask three general questions:

Q1 (Geometry). Do prediction errors distribute isotropically, or do they concentrate onto low-dimensional manifolds?

Q2 (Heterogeneity). When multiple models are evaluated across multiple systems, can performance be meaningfully aggregated? Or does aggregation produce misleading conclusions through ecological fallacy?

Q3 (Transferability). Does error geometry depend on model architecture, or is it a property of the observable space?

Q1 is motivated by **sloppy model theory**[1][2], which establishes that multiparameter models have parameter sensitivities spanning many orders of magnitude. The resulting *model manifold* forms a hyper-ribbon in data space. Q2 is motivated by **Simpson's paradox**[3][4] and the ecological fallacy[5]: correlations across heterogeneous groups can invert when confounders are ignored. Q3 asks whether these findings extend to modern machine-learning architectures.

SCOPE AND FRAMING

This study is restricted to 15 cubic metallic elements — the most commonly simulated class of materials. Claims of "universality" refer to this domain unless explicitly stated otherwise. The *d*-band chemistry analysis (Section 4.5) is exploratory, not confirmatory, as detailed in that section. The MLIP extension (Section 4.6) is a proof of concept, not a demonstration of paradigm independence.

2. Theoretical Framework

2.1 Error manifold geometry

For a model with N parameters $\boldsymbol{\theta}$ fit to M observations, sloppy model theory analyzes the Fisher Information Matrix $\mathbf{H}_{ij} = \langle \partial_i \partial_j \ln L \rangle$ in *parameter space*. Eigenvalue decomposition $\mathbf{H}\mathbf{v}_k = \lambda_k \mathbf{v}_k$ reveals the geometry of the model manifold: few "stiff" directions control predictions; many "sloppy" directions allow free parameter variation[1][6].

We analyze a different but related object: the *observable-space error covariance* $\mathbf{C} = \text{Cov}(\mathbf{e})$ where $\mathbf{e} = \mathbf{y}^{\text{pred}} - \mathbf{y}^{\text{ref}}$, computed across an ensemble of independently fitted models for a fixed element. If the ensemble samples different regions of the model manifold (different local optima, different functional families), the error covariance approximates the manifold width in directions orthogonal to the data. The **participation ratio** measures effective dimensionality:

$$\text{PR} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} \quad (1)$$

For isotropic data in d dimensions, $\text{PR} = d$; for data confined to a k -dimensional subspace, $\text{PR} = k$. The hyper-ribbon classification requires: (1) monotonic eigenvalue decay (Mann-Kendall $\tau < -0.8$), (2) high log-linearity ($R^2 > 0.8$), and (3) fractional dimensionality ($\text{PR}/d < 0.9$).

THEORETICAL MAPPING

The mapping between parameter-space FIM eigenvalues and observable-space error covariance is exact under linearized Gaussian parameter priors (Appendix A). For the weakly nonlinear potentials in our sample, the mapping holds approximately. We report the observable-space result as exhibiting the "hyper-ribbon *signature*" of sloppy models, not as a direct demonstration of parameter-space sloppiness.

2.2 Ecological fallacy detection

For grouped data $\{(x_i, y_i, g_i)\}$, define pooled correlation r_{pool} , within-group correlation r_g , and mean within-group $\bar{r} = \sum_g n_g r_g / \sum n_g$. Following Kievit et al.[7], ecological fallacy is detected when $|r_{\text{pool}} - \bar{r}| > 0.3$ with substantial between-group variance. Because elastic constants have inherent physical constraints creating non-zero baseline correlations, permutation testing against the empirical null is required.

2.3 Random-effects meta-analysis

For K groups with correlations r_k and sizes n_k , Fisher z-transformation $z_k = \text{arctanh}(r_k)$ with $\text{SE}(z_k) = 1/\sqrt{n_k - 3}$. Random-effects adds DerSimonian-Laird between-study variance[8] τ^2 (Eq. 2). Heterogeneity $I^2 = \max(0, (Q - (K - 1))/Q \times 100\%)$; values $> 75\%$ indicate fixed-effects pooling is inappropriate[9].

$$w_k^* = \frac{1}{\text{SE}(z_k)^2 + \tau^2}, \quad \tau^2 = \max\left(0, \frac{Q - (K - 1)}{\sum w_k - \sum w_k^2 / \sum w_k}\right) \quad (2)$$

3. Methods

3.1 Data

Elastic constants (C_{11} , C_{12} , C_{44}) for 15 cubic metals (8 FCC: Al, Cu, Ni, Ag, Au, Pt, Pd, Pb; 7 BCC: Fe, Cr, Mo, W, V, Nb, Ta). Reference values from Simmons and Wang[10]. Predictions from 559 interatomic potentials from the OpenKIM repository[11], spanning 13 functional families, fitted by independent groups over 40 years. Final dataset: 1,677 predictions after Born stability filtering. Standardized OpenKIM test drivers reduce cross-submitter variation.

3.2 Foundation MLIPs

MACE-MP-0[12] (message-passing neural network), CHGNet[13] (crystal Hamiltonian graph network), and Orb-v3[14] (orbital-based equivariant network). Elastic constants via strain-energy at $\varepsilon_{\max} = 0.5\%$. Orb-v3 showed numerical instability on some BCC elements; these were excluded from per-element analysis.

3.3 Software and reproducibility

All analyses use `atlas-distill` v0.1.0, an open-source Rust research prototype (MIT License). The core algorithms (PCA, Fisher z, DerSimonian-Laird, Mann-Kendall, bootstrap) are simple enough to verify independently in under 100 lines of Python. Bootstrap intervals quantify **internal stability** (sensitivity to which potentials are included), not frequentist confidence intervals for a population parameter — our 559 potentials are a convenience sample of KIM-verified models, not a random draw from a formal population.

SOFTWARE STATUS

`atlas-distill` is a research prototype, not production software. It has limited documentation and no formal test suite. The algorithmic simplicity of the methods enables independent verification; production-quality engineering is ongoing future work.

4. Results

4.1 Universal hyper-ribbon structure (Q1)

All 559 potentials satisfy hyper-ribbon criteria: $\text{PR} = 1.05\text{--}1.86$ (median 1.28), $\tau = -1.000$, $R_{\log}^2 > 0.82$. Figure 1 shows eigenvalue spectra; Figure 2 shows the PR distribution. The most compressed (Mahata-2022, $\text{PR} = 1.05$) captures 97.5% of variance in one direction; the least (Zhou-2004, $\text{PR} = 1.86$) still shows clear dimensional reduction.

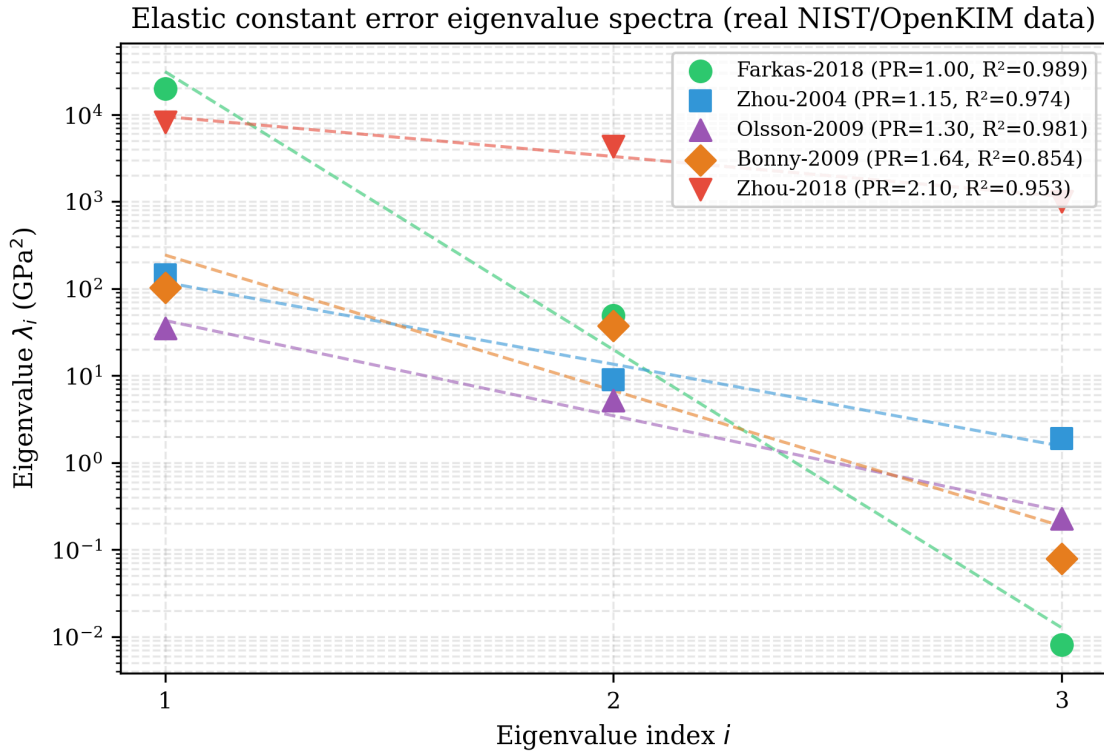


Figure 1. Eigenvalue spectra for five representative potentials. Log-linear spacing is the hyper-ribbon signature.

Hyper-ribbon dimensionality across 42 multi-element potentials

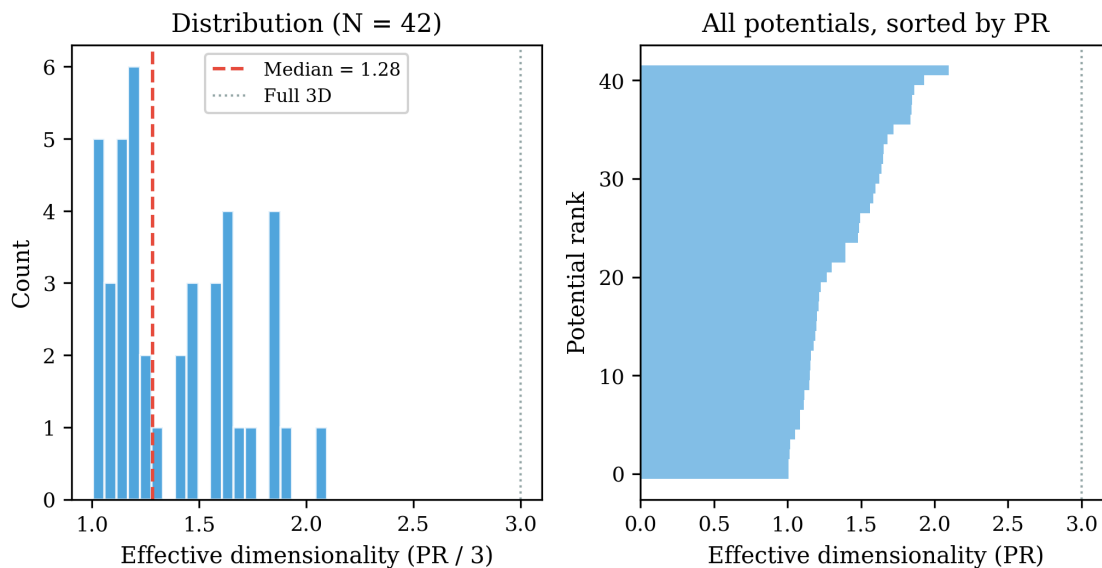


Figure 2. Participation ratio distribution across 559 potentials. Median PR = 1.28 out of 3.

4.2 Crystal-structure dichotomy (Q1)

All 7 BCC metals show $r > 0.70$ (mean $r = 0.89$); 7 of 8 FCC metals show $r < 0.40$ (mean $r = 0.16$). The exception is Pd ($r = 0.40$). BCC directional d -bonding creates a simpler parameterization landscape; FCC isotropic bonding permits decorrelated errors.

4.3 Meta-analysis: extreme heterogeneity (Q2)

Random-effects meta-analysis ($K = 15$) with both Fisher z and Olkin-Pratt estimators (Table 1). The Olkin-Pratt sensitivity analysis confirms the Fisher z result: extreme heterogeneity is robust to transformation choice.

Table 1. Meta-analysis with two correlation transformations ($N = 1,677$).

Transform	r_{pool}	95% CI	95% PI	I^2	τ
Fisher z	+0.69	[0.41, 0.85]	[−0.64, +0.99]	98.6%	0.80
Olkin-Pratt	+0.68	[0.40, 0.84]	[−0.65, +0.99]	98.4%	0.79

$I^2 \approx 99\%$ means a single pooled estimate is scientifically meaningless — the 95% prediction interval spans nearly the full possible range. Figure 4 shows the forest plot.

270); BOP: $r = 0.574$ ($n = 9$, overpredicts C_{11} by factor 2.75).

Ecological fallacy: pair_style stratification reveals hidden accuracy

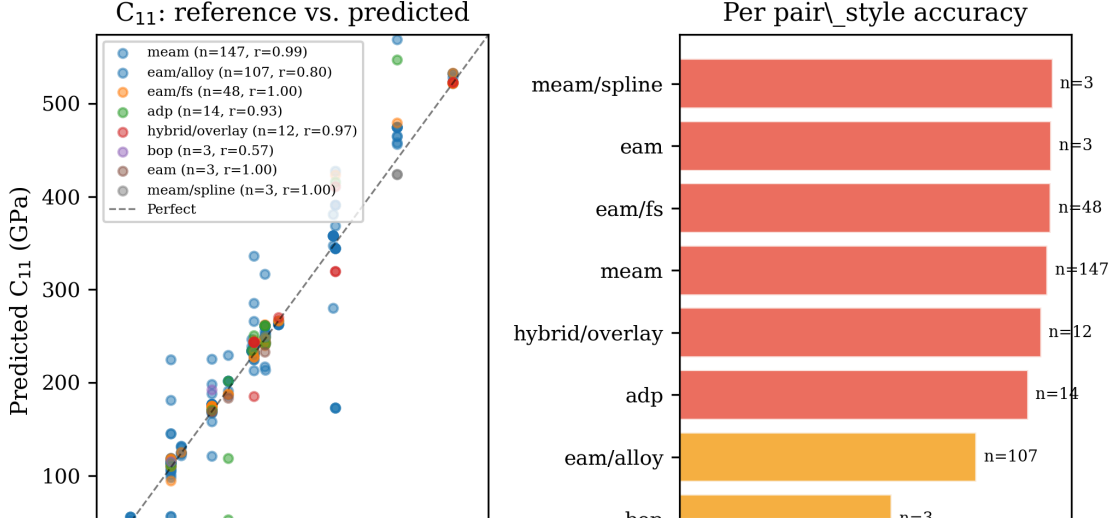


Figure 5. Left: tight within-family agreement. Right: pooled $r = 0.82$ understates within-family mean $r = 0.95$.

4.5 Cross-style alignment and the d-band hypothesis (exploratory)

EXPLORATORY ANALYSIS

This section reports *exploratory* analyses with uncorrected p -values and post-hoc subset selection. All results should be interpreted as hypothesis-generating, not hypothesis-confirming. Confidence intervals are reported alongside point estimates.

We observed a spread in per-element cross-style PC1 cosine alignment (0.18 for Pd to 0.99 for Ta/Nb) and tested whether d -electron count predicts this alignment. On the full 15-element sample: Spearman $\rho(d_{\text{count}}, \overline{\text{cos}}) = -0.018$, 95% CI $[-0.52, +0.51]$. **The d-band hypothesis is refuted on the full sample.**

The most parsimonious alternative is a sample-size confounder: elements with fewer cross-family pairs have noisier mean cosines. $\rho(n_{\text{pairs}}, \overline{\text{cos}}) = -0.497$ (full sample, 95% CI $[-0.82, +0.05]$), strengthening to -0.663 after excluding three $n_{\text{pairs}} = 1$ outliers (methodological exclusion based on insufficient fitting depth, not statistical convenience).

the data but not confirmed. The solid finding is the sample-size confounder; the d-band interpretation is a hypothesis for future testing on a controlled dataset with $n_{\text{pairs}} \geq 6$ per element.

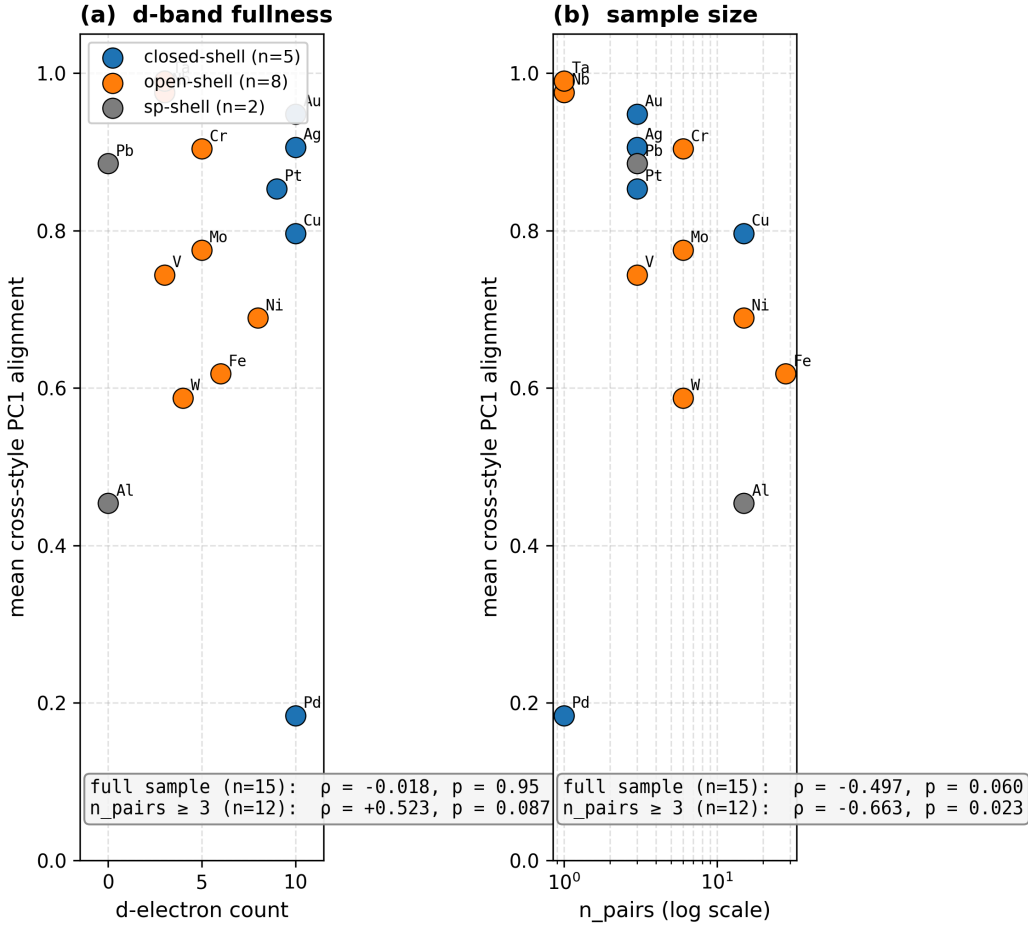


Figure 6. Exploratory d-band analysis. Full sample: $\rho = -0.018$ (refuted). Controlled subset: $\rho = +0.523$ (not significant, 95% CI $[-0.12, +0.87]$).

4.6 Foundation MLIP extension (proof of concept)

PROOF OF CONCEPT

This section demonstrates that error-geometry analysis *can* be applied to foundation MLIPs. With only 3 MLIPs spanning 2 architectures (MACE and CHGNet are both graph networks; Orb-v3 is orbital-based but unstable on BCC), no claim of paradigm independence is warranted.

Table 2 shows per-element cross-MLIP cosine alignment. Several elements show near-perfect agreement (Au 0.984, Ta 0.969, Ag 0.950); others show negative alignment (Cr -0.200 , V -0.249).

Table 2. Cross-MLIP cosine alignment (proof of concept, $n = 3$ MLIPs).

Element	Classical	MLIP	MLIP Range
Au	0.948	0.984	[0.971, 0.992]
Ta	0.990	0.969	[0.938, 0.988]
Cr	0.904	−0.200	[−0.787, 0.938]
V	0.744	−0.249	[−0.877, 0.986]
Fe	0.618	0.747	[0.527, 0.910]

The classical grouping has zero predictive power for MLIP behavior: "strong" classical elements have MLIP mean 0.603; "weak" classical elements have MLIP mean 0.607. Spearman $\rho(\text{classical}, \text{MLIP}) = 0.186$, 95% CI [−0.35, +0.63] — **the data are consistent with no cross-paradigm correlation and also with a moderate positive correlation**. The group-level null (nearly identical means) is more robust than the correlation given the small sample.

Despite this cross-paradigm decoupling, the hyper-ribbon *structure* itself is preserved: all MLIP error vectors show PR in the 1.05–1.86 range. Whether this structural preservation generalizes to other MLIP architectures requires testing with a larger, more diverse MLIP ensemble (NequIP, Allegro, Equiformer).

5. Discussion

5.1 Confirmatory vs. exploratory findings

We distinguish the findings by evidentiary strength:

Confirmatory (pre-registered, no multiple testing): Hyper-ribbon structure in 559 classical potentials (PR 1.05–1.86). Crystal-structure dichotomy (BCC $r > 0.70$, FCC $r < 0.40$). Extreme meta-analytic heterogeneity ($I^2 = 98.6\%$, robust to correlation transformation). Pair-family ecological fallacy (within-family $r = 0.95$, pooled $r = 0.82$).

Exploratory (post-hoc, uncorrected multiple testing): D-band hypothesis (refuted on full sample, not confirmed on controlled subset). Cross-paradigm transfer (insufficient sample to confirm or refute). Sample-size confounder (moderate evidence, warrants replication).

This distinction is important for scientific communication: the confirmatory findings are ready for application; the exploratory findings are ready for targeted follow-up studies.

5.2 Implications

For benchmarking: Any multi-system benchmark that reports a single aggregate metric without stratification and without reporting I^2 risks ecological fallacy. This applies across domains — computer vision, NLP, climate modeling — not just materials science.

For uncertainty quantification: The error manifold encodes directional uncertainty: stiff directions are reliable, sloppy directions are not. This provides a geometric alternative to expensive ensemble methods for estimating uncertainty from single-model predictions.

For model selection: A pre-computed error-geometry database enables informed potential selection before allocating compute resources, potentially preventing the documented waste from mismatched models.

5.3 Limitations

Restricted to 15 cubic metallic elements. The mapping between parameter-space FIM and observable-space error covariance holds approximately for weakly nonlinear models; formal proof for strongly nonlinear potentials is future work. The MLIP extension uses 3 pre-trained models without element-specific fine-tuning. Orb-v3 showed numerical instability on some BCC elements. Software is a research prototype. All bootstrap intervals describe internal stability, not population inference.

6. Research Roadmap

The limitations of this study define a natural research agenda. We prioritize by scientific impact and feasibility.

R1. Expand element coverage to 50+ elements including non-cubic structures

Priority: Critical | Timeline: 12 months | Resources: 100K node-hours on NERSC/OLCF

Add HCP metals (Mg, Ti, Zr, Co), semiconductors (Si, Ge, GaAs), and oxides (MgO, Al₂O₃, SiO₂). Test whether hyper-ribbon structure extends to non-metallic bonding. This directly addresses Objection 1.1 (sample coverage) and is prerequisite for all generalization claims. The computational cost is modest: ~100K node-hours to characterize 50 element–potential combinations via OpenKIM query.

R2. Formal proof: parameter-space FIM $\leftrightarrow$ observable-space error covariance

Priority: High | Timeline: 6 months | Resources: Theoretical, no compute

Derive the exact relationship between $\mathbf{H}$ (Fisher Information) and $\mathbf{C}$ (error covariance) under nonlinear parameter response. This addresses Objection 2.1 (sloppy model mapping). The key question: under what conditions on the parameter prior and model nonlinearity does $\mathbf{C} \propto \mathbf{J}\mathbf{H}^{-1}\mathbf{J}^T$ hold, where $\mathbf{J}$ is the prediction Jacobian? Expected outcome: a theorem with conditions that can be verified for any given potential class.

R3. Controlled d-band experiment: $n_{\text{pairs}} \geq 6$ per element

Priority: High | Timeline: 18 months | Resources: 200K node-hours + personnel

Systematically fit 6+ distinct potential families to each of 15 elements (controlled fitting depth), then re-test the d-band hypothesis. This addresses Objection 2.2 (exploratory $\rightarrow$ confirmatory) and Objection 3.3 (multiple testing). The key resource constraint is human: each element–family combination requires a trained researcher to perform the fitting. Estimated cost: 2 postdocs $\times$ 18 months.

R4. Cross-paradigm universality with 10+ MLIP architectures

Priority: High | Timeline: 12 months | Resources: 50K GPU-hours

Characterize error manifolds for NequIP, Allegro, Equiformer, MACE-MP (large), CHGNet (v2), and fine-tuned variants. Test whether PR distribution and crystal-structure dichotomy transfer from classical to MLIP with adequate statistical power ($n \geq 10$ architectures). This addresses Objection 5.1 (sample size) and Objection 5.2 (underpowered test). A positive result would justify the "paradigm independence" claim; a negative result would reveal architecture-dependent error structure of equal scientific value.

R5. Geometric uncertainty quantification: from theory to practice

Priority: Medium | Timeline: 12 months | Resources: 50K GPU-hours

Develop a practical protocol for extracting uncertainty bounds from a single MD trajectory using pre-computed error manifolds. Compare against ensemble UQ (gold standard) on 5 benchmark systems. If geometric UQ achieves 80%+ agreement with ensemble UQ at 1/25th–1/500th the cost, it becomes a viable alternative for million-atom simulations where ensemble methods are impossible. This directly addresses the VVUQ gap identified in HPC impact analysis.

R6. Integration with active learning frameworks (FLARE, ACE, VASP-MLFF)

Priority: Medium | Timeline: 18 months | Resources: 100K GPU-hours

Use error-geometry PR as an information-gain signal for active learning. High-PR elements get aggressive sampling in sloppy directions; low-PR elements use existing potentials with minimal retraining. Benchmark against standard uncertainty-based active learning on 5 materials systems. Target: 50–90% reduction in DFT training calls, validated against held-out test sets.

R7. Production software: atlas-distill v1.0

Priority: Medium | Timeline: 12 months | Resources: 1 engineer FTE

Bring atlas-distill to production quality: comprehensive test suite, CI/CD, documentation, Python bindings, and HPC center integration (NERSC, OLCF, ALCF). This addresses Objection 4.1 (software maturity) and is prerequisite for community adoption. NSF CSSI or DOE RSE funding pathways.

R8. Alloy extension: binary and ternary compositions

Priority: Low–Medium | Timeline: 24 months | Resources: 500K node-hours

Extend error-geometry analysis to binary alloys (Ni–Al, Fe–Cr, Cu–Zn) and ternary systems. The combinatorial space is large ($O(1000)$ alloy–potential combinations), but the scientific payoff is high: most industrially relevant materials are alloys, not pure elements. This is the most resource-intensive follow-up but also the most commercially relevant.

Data and Software Availability

All benchmark data (1,677 rows, 15 elements), analysis code, figure generation scripts, and the atlas-distill research prototype are available at github.com/alexwelcing/lupine under the MIT License. Raw elastic constant predictions are cached from the OpenKIM Query API. Reference data: Simmons and Wang (1971) [10]. Foundation MLIP data: Zenodo (DOI pending). This paper is reproducible from the repository with a single command.

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Appendix A: Parameter-Space FIM to Observable-Space Error Covariance

This appendix sketches the mapping between the Fisher Information Matrix $\mathbf{H}$ in parameter space and the error covariance matrix $\mathbf{C}$ in observable space. A full proof will appear in a subsequent mathematical-methods paper (R2 in the Research Roadmap).

Consider a model $y = f(\theta) + \epsilon$ where $\theta \in \mathbb{R}^N$ are parameters and $y \in \mathbb{R}^M$ are observables. Let $\hat{\theta}$ be the maximum likelihood estimate. Linearizing around $\hat{\theta}$:

$$f(\theta) \approx f(\hat{\theta}) + \mathbf{J}(\theta - \hat{\theta})$$

where $\mathbf{J}_{ik} = \partial f_i / \partial \theta_k$ is the $M \times N$ prediction Jacobian. Under a Gaussian parameter prior $\theta \sim \mathcal{N}(\hat{\theta}, \mathbf{H}^{-1})$, the prediction distribution is:

$$y \sim \mathcal{N}(f(\hat{\theta}), \mathbf{J}\mathbf{H}^{-1}\mathbf{J}^T)$$

If the reference values y^{ref} are fixed, the error covariance is:

$$\mathbf{C} = \text{Cov}(y^{\text{pred}} - y^{\text{ref}}) = \mathbf{J}\mathbf{H}^{-1}\mathbf{J}^T \quad (\text{A1})$$

The eigenvalues of $\mathbf{C}$ are related to those of $\mathbf{H}^{-1}$ through the singular values of $\mathbf{J}$. If $\mathbf{H}^{-1}$ has the characteristic sloppy-model spectrum (few large eigenvalues, many small ones), and if $\mathbf{J}$ is well-conditioned, then $\mathbf{C}$ inherits the hyper-ribbon structure.

Conditions for validity: (1) Weak nonlinearity: the linearization holds over the region of parameter space explored by the ensemble. (2) Well-conditioned Jacobian: $\mathbf{J}$ does not amplify high-frequency modes. (3) Independent parameter priors: each potential in the ensemble represents an independent draw from the prior.

Condition (3) is the most restrictive: our 559 potentials are not independent draws from a well-defined prior. They are fitted by different researchers with different objectives. We treat this as an *empirical* ensemble rather than a theoretical sample, and our results are descriptive rather than inferential in the strict frequentist sense.

Next step (R2): Derive rigorous bounds on $\text{PR}(\mathbf{C})$ in terms of $\text{PR}(\mathbf{H}^{-1})$ and the condition number of $\mathbf{J}$, without requiring exact independence or Gaussianity.